



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 7-6 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Solve

Graph

Solution

Substitute

Solution set

Inverse operation

Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1 To find all the values that make an inequality true is to it.

2 To show the solutions on a number line is to them.

3 A value that makes an inequality true is a .

4 To check a solution, I the value back into the inequality.

5 All the values that make an inequality true form the .

6 An operation that undoes another to isolate the variable is an .

Reflect — Exit Ticket

Solve and describe the graph: $x + 5 < 11$

- A. $x < 6$; open circle at 6, shade left
- B. $x < 16$; open circle at 16, shade left
- C. $x \leq 6$; closed circle at 6, shade left
- D. $x > 6$; open circle at 6, shade right

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation
7. **Watch for this mistake:** *A common mistake in Solve and Graph Inequalities is skipping the key idea: "Solve with inverse operations, then graph the solution set: open circle for $<$ or $>$, closed for $\leq$ or $\geq$." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Yes, because $5 \leq 5$ is true — *Substitute $x = 5$ into $x \leq 5$: $5 \leq 5$ is true because $\leq$ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.*
9. **Try It 2:** A. All numbers greater than 7, like 8, 9, 10, ... (but NOT 7 itself) — *The solution set is every value that makes $x > 7$ true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because $>$ does not include the boundary, 7 itself is NOT in the solution set.*
10. **Exit Ticket:** A. $x < 6$; open circle at 6, shade left — *Subtract 5 from both sides: $x < 11 - 5 = 6$. Open circle at 6 (not included), shade left. ✓*